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1 Appendix E: Spectral Unmixing and Classification

1.1 Objective

Comprehensive spectral analysis including advanced unmixing algorithms, feature extraction, and multi-algorithm classification for CHC identification and chemical analysis.

1.2 Methods (src)

- `src/case_studies/spectral_unmixing.py`
 - `generate_realistic_chc_spectra(n_compounds, n_wavelengths)` - Synthetic CHC spectral data
 - `nmf_unmix(spectra, n_components, seed=42)` - Non-negative matrix factorization
 - `vertex_component_analysis(spectra, n_endmembers)` - VCA endmember extraction
 - `independent_component_analysis_spectra(spectra, n_components)` - ICA blind separation
 - `spectral_feature_extraction(spectra, method)` - Multi-method feature extraction
 - `advanced_classification_suite(features, labels)` - Multi-algorithm classification
 - `performance_metrics_comprehensive(y_true, y_pred, y_prob)` - Detailed evaluation metrics
 - `lda_baseline(features, labels, seed=42)` - Linear discriminant baseline

1.3 Script and Outputs

- Script: `scripts/generate_spectral_unmixing.py`
- Data: `output/data/spectral_unmixing_comprehensive.npz`
- Figure: `output/figures/spectral_unmixing_comprehensive_analysis.png`

1.4 Figure

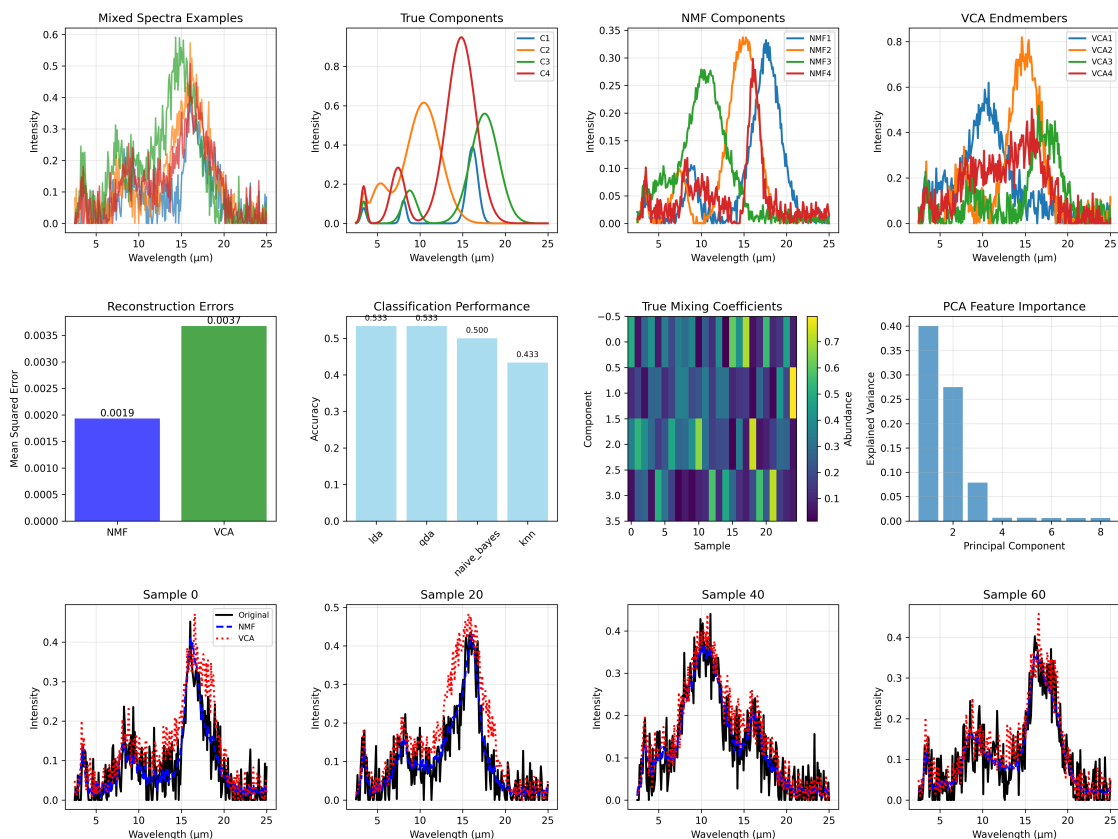


Figure 1: Comprehensive spectral analysis: Synthetic CHC spectra with realistic chemical diversity, advanced unmixing using NMF, VCA, and ICA algorithms, multi-method feature extraction (peaks, derivatives, PCA, statistical), and multi-algorithm classification performance comparison. Includes detailed performance metrics, confusion matrices, and algorithmic benchmarking for chemical identification applications.

1.5 Equation References

- Spectral overlap: see (??) analogs for information metrics; overlap in main text.

1.6 Reproducibility

- Run: `python3 scripts/generate_spectral_unmixing.py`
- Artifacts saved to `output/data/` and `output/figures/`.
- Fixed RNG seed (42) used for NMF initialization and fold splits.

1.7 Cross-References

- Methods: ??
- Symbols: ??
- Math appendix: ??